ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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Abstract. Vision–language models (VLMs) such as Vintern-1B have brought strong multimodal abilities to Vietnamese, but adapting them to a new Visual Question Answering (VQA) benchmark remains costly: the usual recipe fine-tunes the language model on many high-resolution image tiles, while task-specific architectures must be trained from scratch. In this paper, we propose **ViBridge-VQA**, a parameter-efficient approach that improves Vintern-1B without updating any of its pre-trained weights. ViBridge-VQA feeds the frozen language decoder two complementary kinds of visual tokens from a single image view: *global* tokens, produced by a small trainable bridge from the `[CLS]` vector that Vintern-1B discards, and *local* tokens, produced by Vintern-1B's own pre-aligned projector and pooled to a chosen budget. We treat the number of tokens as an accuracy–cost trade-off and select it on validation data as the knee of the Pareto front. On the reasoning-oriented AutoViVQA benchmark, the selected budget of 50 visual tokens gives two main results. Training only the bridge (1.35% of the parameters) raises the F1 of Vintern-1B from 17.6 to 54.6, above the published fine-tuned Vintern-1B, which reads up to 36 times more visual tokens. Training a lightweight decoder LoRA together with the bridge (1.57% of the parameters) raises F1 further to 58.2. Local tokens matter far more than global ones, and they also transfer better to unseen Vietnamese VQA datasets than decoder adaptation alone.

Keywords: Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. The task has progressed rapidly with large vision–language models (VLMs) such as BLIP-2 [11], Flamingo [1] and LLaVA [17],

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which connect a pre-trained vision encoder to a pre-trained language model and generate free-form answers. Vietnamese has followed this trend: Vintern-1B [6] connects an InternViT-300M encoder [5] to a Qwen2.5-0.5B decoder [25] and is among the strongest open Vietnamese VLMs, while benchmarks such as ViVQA [27], OpenViVQA [20] and the recent reasoning-oriented AutoViVQA [19] make it possible to measure progress in this low-resource setting.

Obtaining a strong model on a new Vietnamese benchmark, however, is still expensive. The common recipe fine-tunes an existing VLM: the fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model and splits every image into up to six high-resolution tiles plus a thumbnail, so that the decoder reads up to 1,792 visual tokens per image. The alternative is to design a task-specific architecture, as ViMoE-VQA [18] does with a sparse mixture-of-experts fusion module and a decoder trained on AutoViVQA. A cheaper option is to keep the whole VLM frozen and train only a small connector between its encoder and decoder. Existing connectors follow two designs, each with its own weakness. A *global-only bridge* maps the global image vector to a few tokens; it is cheap, but it loses local details such as small objects, printed text and spatial relations. A *learned patch compressor*, such as a Q-Former, keeps these details, but it has to learn a new mapping from visual features to the decoder, which is difficult with limited Vietnamese data.

Vintern-1B suggests a third option. Its projector was pre-trained to map image regions into the input space of the decoder, so this alignment already exists and does not need to be learned again. At the same time, the model discards the global `[CLS]` vector of its encoder, which summarises the whole scene. This raises our research question: *when labelled data is limited, is reusing the pre-trained visual alignment of a VLM, with pooling to control the number of tokens, more effective than learning a new connector, and how many global and local tokens does the frozen decoder need?*

To address this question, we propose **ViBridge-VQA**, which gives the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector of Vintern-1B, average-pooled to a chosen budget and placed in its original image slot, so the decoder receives region features in the representation it was trained to read. *Global* tokens are produced from the `[CLS]` vector by a small trainable bridge and add the scene-level information that Vintern-1B ignores. Only this bridge is trained. Because every additional token increases inference cost, we treat the number of global and local tokens as a budget and select it as the knee of the accuracy–cost Pareto front. As a result, ViBridge-VQA with 14 global and 36 local tokens exceeds the published results of the fine-tuned Vintern-1B on most metrics while reading up to 36 times fewer visual tokens and updating none of its weights.

In summary, our main contributions are as follows:

- We propose **ViBridge-VQA**, a global–local bridge that improves a frozen Vietnamese VLM by combining trainable global tokens with local tokens from its own pre-trained projector. With 50 visual tokens, it raises the F1 of Vintern-1B on AutoViVQA from 17.6 (zero-shot) to 54.6 when only the

bridge is trained (1.35% of the parameters), and to 58.2 when a lightweight decoder LoRA is trained with it (1.57%).
- We study the **visual token budget** systematically over eight configurations and three random seeds, and select it on the validation split as the knee of the accuracy–cost Pareto front. The choice is the same for every metric and for cost measured in tokens or in runtime.
- We **analyse where the gains come from**: local tokens matter far more than global ones, work better from the pre-trained projector than from a learned compressor, complement decoder adaptation, and transfer to four unseen datasets.

2 Related Work

Early VQA systems classified over a fixed set of answers, whereas current generative VLMs produce free-form answers with an autoregressive language model [11,1,17]. Vietnamese datasets include ViVQA [27], OpenViVQA [20], ViCLEVR [29], ViOCRVQA [24] and ViTextVQA [22]. AutoViVQA [19] contributes 37,077 questions on 19,411 COCO images [15], generated by large language models, filtered by an ensemble of validators and annotated with reasoning categories, with an emphasis on relational, causal and multi-step reasoning. On the modelling side, BARTPhoBEiT [28] and AViVQA-TranConI [21] combine pre-trained visual and Vietnamese language representations, Vintern-1B [6] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [18] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. These approaches either train a new model or fine-tune an existing VLM on multi-tile inputs; in contrast, we keep every pre-trained weight of Vintern-1B frozen and use a single image view.

Keeping the vision encoder and the language model frozen and training only a connector is a common way to build VLMs efficiently, and connectors differ mainly in what they must learn. Frozen [30] learns a visual prefix from a global image representation. The Q-Former of BLIP-2 [11] learns a set of queries that compress the patch features, which preserves details but requires a large module (188M parameters) pre-trained on millions of image–text pairs. LLaVA [17,16] projects every patch token with a small MLP, which preserves locality but passes hundreds of tokens to the decoder; Honeybee [4] proposes locality-preserving abstractors, and InternVL [5], on which Vintern-1B is built, merges neighbouring patches before its MLP projector. In all of these models, the mapping from patches to the decoder is learned during large-scale pre-training. ViBridge-VQA differs in two respects: it does not learn this local mapping again but reuses the projector already pre-trained in the VLM, and it trains only a small global bridge (12.9M parameters) on about 26k task questions.

The number of visual tokens largely determines the cost of the decoder. Inference-optimal scaling of VLMs [12] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] train a model to work at several nested pooled token budgets and show that the required budget depends on the task. These approaches train or re-train the model for each budget. In

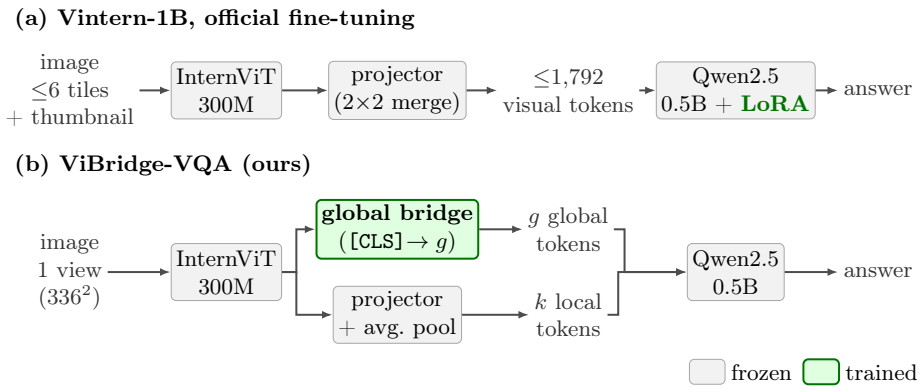


Fig. 1: Overview of ViBridge-VQA compared with the official fine-tuning of Vintern-1B. (a) The official recipe encodes up to seven views per image, passes up to 1,792 visual tokens to the decoder and adapts the decoder with LoRA. (b) ViBridge-VQA encodes a single view and passes g global tokens, produced from the `[CLS]` vector by a trainable bridge, and k local tokens, produced by the frozen projector of Vintern-1B and average-pooled. Only the global bridge is trained (1.35% of the parameters for $g = 14$); the selected budget is $g = 14$, $k = 36$.

contrast, we keep the model frozen and obtain each budget simply by pooling the output of the frozen projector, and we choose among budgets with explicit criteria, namely the knee of the Pareto front [26] and TOPSIS [10]. Parameter-efficient methods such as adapters [8], prefix tuning [13] and LoRA [9] reduce the cost of adaptation from the other side, by training a small number of additional weights in the decoder. We use decoder LoRA both as a baseline and as a complement, and ask whether a small training budget is better spent on the visual tokens or on adapting the decoder.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q, the model generates a short free-form answer $y = (y_1, \ldots, y_T)$. ViBridge-VQA builds on Vintern-1B [6] and keeps all of its pre-trained weights frozen. Instead of adapting the model, it changes what the decoder sees.

3.1 Overall Architecture

As illustrated in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector P, and the Qwen2.5-0.5B-Instruct language decoder (24 layers, width 896). Each image is resized to one 336×336 view (24×24 patches of 14×14 pixels); the encoder returns one vector per patch and a global `[CLS]` vector

$v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: it merges every 2×2 block of neighbouring patches into one 4096-dimensional vector (pixel shuffle) and maps each merged vector into the decoder input space with P, which was pre-trained to align the two backbones. At our resolution this yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

3.2 Global Tokens

The `[CLS]` vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge g_ϕ expands it into g tokens with two linear heads:

$$g_\phi(v) = \big[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\big] \in \mathbb{R}^{g \times 896}. \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head produces an *anchor* token, and the second produces $g-1$ complementary tokens that present the global content to the decoder from different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters, i.e., 12,857,600 for $g = 14$, and its g tokens are placed before the prompt. We use $g = 14$ because, in a preliminary study, the F1 of this bridge alone rose with g up to 14 and then plateaued.

3.3 Local Tokens

Global tokens alone cannot describe small objects, printed text or the position of objects. ViBridge-VQA therefore also passes local tokens, taken directly from the 144 region tokens of the frozen projector P. To control their number, we average-pool the 12×12 grid to an $s \times s$ grid, so that $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each 2×2 block. The pooled tokens replace the embeddings of k image-context placeholders inside Vintern-1B's own image slot, `<img>` $\cdots$ `</img>`, in the prompt. Because they come from the pre-trained projector rather than from a newly learned projection of the patches, the local tokens stay in the representation that the decoder was trained to read, and no parameters are needed to produce them.

3.4 Training Objective

Only the bridge parameters ϕ are trained; the vision encoder, the projector and the decoder remain frozen, and no adapter is added to the decoder. We minimise the cross-entropy of the answer tokens:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, g_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where x denotes the merged patch vectors. ViBridge-VQA can also be combined with decoder adaptation: in that variant, a rank-16 LoRA [9] on the query, key, value and output projections of every decoder layer is trained jointly with the

bridge, which adds 2,162,688 parameters. Throughout the paper, percentages of trained parameters are relative to all parameters of the model, i.e., the frozen components of Vintern-1B (about 0.94B, including the vision encoder, projector, embeddings and output layer) plus the trained modules. With $g = 14$, the bridge alone trains 12.86M parameters (1.35%), and the bridge with LoRA 15.02M (1.57%).

3.5 Choosing the Token Budget

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair (g, k) as a budget to be chosen. In our experiments, accuracy increases monotonically with the number of visual tokens, so every configuration lies on the accuracy–cost Pareto front and the front alone does not single one out. We therefore select its *knee* [26], the point after which additional tokens bring diminishing returns. After normalising cost c and accuracy a to $[0, 1]$ over the front, the knee is the configuration that maximises $\hat{a} - \hat{c}$, i.e., the one lying furthest above the line joining the cheapest and the most accurate configuration. The knee is computed on the validation split only, and we check the choice with the multi-criteria method TOPSIS [10], also on validation, under several weightings of accuracy and cost. Test results are computed for every configuration only to be reported; no choice is based on them.

4 Experiments

4.1 Dataset and Evaluation Metrics

Dataset. We evaluate on AutoViVQA [19], a Vietnamese VQA benchmark built on MS COCO images [15] with an LLM-driven generation pipeline and ensemble-based quality control. Each question has five short free-form reference answers of 1–10 words and is labelled with its reasoning type. In the public 80/20 split of the release, 2,941 of the 7,416 validation questions concern images that also appear in training. To avoid this overlap, we remove 273 duplicated (image, question) pairs and 97 questions with irregular type labels, keep the eight remaining reasoning categories, and re-partition the 36,707 questions by image into a 70/15/15 split (seed 42). This gives 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images. All our results are reported on this grouped split; in a preliminary comparison, it lowered the F1 of the global-only bridge by about 0.5 points relative to a split that does not group questions by image.

Evaluation metrics. Following AutoViVQA and ViMoE-VQA, we report answer-matching metrics (exact-match Accuracy, and word-level Precision, Recall and F1) and generation metrics (BLEU [23], ROUGE-L [14], METEOR [2] and CIDEr [31]), all multiplied by 100. Word-level precision, recall and F1 are computed against each of the five references, and the reference with the highest F1 is kept for all three. All our numbers are produced by one implementation

of these metrics. We average F1 over questions, which gives lower values than the harmonic mean of averaged precision and recall used for the published baselines (55.95 instead of 54.56 for ViBridge-VQA), so our F1 comparisons are conservative.

4.2 Experimental Setup

Implementation details. The vision encoder, the projector and the decoder are initialised from Vintern-1B-v3.5 [6] and kept frozen. ViBridge-VQA is trained for two epochs, after which neither the validation loss nor CIDEr improves further; the variants with a decoder LoRA are trained for one or three epochs. We use AdamW with a learning rate of 2×10^{-4} and an effective batch size of 8, train on the first reference answer, and decode greedily (batch size 2, at most 50 new tokens). Unless stated otherwise, every configuration is trained with three random seeds (42, 123 and 3407), and we report the mean and the standard deviation. All runs fit on a single 16 GB GPU (Tesla P100 or T4). The token budget is selected on the validation split, and the test split is used only for the final report.

Baselines. We compare against three groups of systems: (1) multimodal models fine-tuned on AutoViVQA, namely ViT5_ViT, BARTPhoBEiT [28], Vintern-1B and ViMoE-VQA [18]; (2) models used zero-shot, namely Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5; and (3) our *global-only* model, which uses the same bridge without local tokens ($k = 0$, $g = 8$), with and without a decoder LoRA. The results of groups (1) and (2) are taken from the AutoViVQA and ViMoE-VQA papers and were obtained on splits that differ from ours (ViMoE-VQA reports an 80/10/10 split). We report them as published reference points. The fine-tuned Vintern-1B follows the official recipe: the vision encoder and the projector are frozen, a rank-16 LoRA adapts the language model for one epoch, and each image is encoded as up to six 448×448 tiles plus a thumbnail.

4.3 Main Results

Table 1 compares ViBridge-VQA with the baselines. Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 in the zero-shot setting to 54.56, and its CIDEr from 8.54 to 110.58. With the selected budget of 50 visual tokens, it already exceeds the published scores of the fine-tuned Vintern-1B on seven of the eight metrics; the largest gains appear in the generation metrics (BLEU, METEOR and CIDEr), and only exact-match accuracy remains 1.3 points lower. When all 144 local tokens are used, it exceeds them on every metric, including accuracy, while still reading up to eleven times fewer visual tokens. These results suggest that much of what AutoViVQA requires is already present in the frozen Vintern-1B, provided that its decoder receives suitable visual input.

Decoder adaptation adds to these gains rather than replacing them. Training ViBridge-VQA jointly with a rank-16 LoRA for a single epoch raises test F1 by a further 3.6 points to 58.17 and gives the best accuracy, BLEU, ROUGE-L, METEOR and CIDEr in Table 1 (setting aside the outlying BARTPhoBEiT

Table 1: Performance on the AutoViVQA test set (×100). Baseline results are those published in AutoViVQA [19] and ViMoE-VQA [18] (ViMoE-VQA: 80/10/10 split, mean of five seeds); our models use the grouped split of Section 4.1. For our models we report the mean ± standard deviation over three seeds (global only: four), the share of trained parameters, and use 8 (global only), 50 (k=36) or 158 (k=144) visual tokens, against up to 1,792 for the fine-tuned Vintern-1B. Best results per column are in bold. †Far outside the range of all other systems and not used for comparison.

Category	Model	Acc	Prec	Rec	F1	BLEU	ROUGE-L	METEOR	CIDEr
Multimodal (fine-tuned)	ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
	BARTPhoBEiT	8.81	45.30	46.48	45.88	43.29†	44.83	24.57	188.96†
	Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
	ViMoE-VQA	9.55	**62.89**	58.60	**60.69**	12.54	47.12	39.10	88.76
LLMs and VLMs (zero-shot)	Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
	LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
	Gemini 2.0 Flash	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
	Gemini 2.5 Flash	0.22	24.43	**76.66**	24.75	0.39	37.27	31.22	71.90
	GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Global only (ours, k=0)	g=8 (0.78%)	8.86 ±.37	51.19 ±.19	52.37 ±.21	50.47 ±.22	16.24 ±.43	48.70 ±.19	41.10 ±.29	96.19 ±1.10
	+ LoRA, 3 ep. (1.01%)	11.40 ±.24	55.37 ±.18	56.16 ±.09	54.52 ±.14	20.97 ±.29	52.68 ±.21	45.08 ±.14	108.43 ±.67
ViBridge-VQA (ours, g=14)	k=36 (1.35%)	11.75 ±.21	55.36 ±.21	56.56 ±.21	54.56 ±.19	18.52 ±.18	52.64 ±.19	45.53 ±.33	110.58 ±.95
	k=144 (1.35%)	13.19 ±.23	56.64 ±.13	58.17 ±.26	56.04 ±.19	19.79 ±.15	54.11 ±.16	47.38 ±.26	116.74 ±.76
	k=36 + LoRA, 1 ep. (1.57%)	**14.67** ±.30	59.02 ±.12	59.96 ±.33	58.17 ±.19	**22.53** ±.11	**56.25** ±.22	**49.32** ±.37	**122.88** ±.93

values). The task-specific ViMoE-VQA, whose fusion module and decoder are trained on AutoViVQA, keeps the highest precision and F1, 2.5 F1 points above our best model. ViBridge-VQA reaches its results differently: instead of designing and training a new architecture, it reuses an existing VLM and trains 1.57% of its parameters, and it leads on accuracy and on all generation metrics. Zero-shot language models reach a high recall with long answers but lose precision, whereas our answers match the reference length (Section 4.6).

Our global-only model isolates the effect of the local tokens: even with a frozen decoder, ViBridge-VQA matches the global-only model trained with a three-epoch decoder LoRA in F1 (54.56 versus 54.52) and exceeds it in CIDEr (110.58 versus 108.43), although its BLEU is lower.

Statistical stability. On the test split, the standard deviation of ViBridge-VQA is at most 0.19 for F1 and 0.95 for CIDEr, and for each of our models in Table 1, test F1 lies at most 0.6 points below validation F1.

4.4 Choosing the Token Budget

Table 2 and Fig. 2a vary the number of global tokens $g \in \{8, 14\}$ and of local tokens $k \in \{1, 9, 36, 144\}$. The most striking pattern is that *local tokens matter far more than global ones.* At $g = 14$, increasing k from 1 to 144 adds 4.9 F1 and 19.1 CIDEr on the validation split, whereas increasing g from 8 to 14 at a fixed k adds only 0.2–0.6 F1. Accuracy has not saturated within the 144 region tokens available at our resolution, as each step of k still adds 1.3–2.0 F1, but the gain per token drops quickly, from 0.07 F1 per token between 23 and 50 tokens to 0.015 between 50 and 158. The budget is therefore a genuine trade-off between accuracy and cost.

Table 2: Effect of the token budget (×100, mean over three seeds, ± standard deviation for F1; global only: four seeds). Tokens: visual tokens passed to the decoder ($g + k$). Time: evaluation runtime (minutes) of one session that loads the model and answers the 10,931 validation and test questions (greedy, batch 2), mean over seeds (Kaggle P100/T4). The selected budget is in bold.

				Validation		Test	
g	k	Tokens	Time	F1	CIDEr	F1	CIDEr
8	0	8	–	50.74 ±.17	99.07	50.47 ±.22	96.19
8	1	9	57.0	51.03 ±.12	99.27	50.71 ±.16	96.90
8	9	17	59.2	52.82 ±.11	106.11	52.11 ±.25	101.30
8	36	44	60.6	54.32 ±.41	111.88	53.82 ±.22	107.67
8	144	152	71.2	56.15 ±.08	119.05	55.20 ±.07	114.13
14	1	15	58.8	51.67 ±.25	101.66	51.67 ±.12	99.53
14	9	23	57.7	52.98 ±.08	106.02	52.76 ±.28	103.63
14	**36**	**50**	**60.0**	**54.96** ±.28	**113.29**	**54.56** ±.19	**110.58**
14	144	158	70.2	56.61 ±.04	120.71	56.04 ±.19	116.74

Because accuracy grows with every added token, all configurations lie on the Pareto front, and we select its knee (Section 3.5). On the validation split, the knee is $g = 14$, $k = 36$, and it is the same for each of the eight metrics, whether cost is counted in visual tokens or as evaluation runtime; As a post-hoc check that played no role in the selection, the knee computed on the test split falls on the same configuration. On validation, TOPSIS (benefits: gains in F1, CIDEr and METEOR; costs: tokens, runtime, trained parameters, seed variance) makes the same choice when accuracy and cost are weighted equally, and $g = 14$, $k = 36$ is the only configuration in its top four for every accuracy weight between 0.3 and 0.7. In practical terms, it obtains 72% of the F1 gain of $k = 144$ over the global-only model with 32% of its visual tokens, and it is the cheapest configuration that reaches the global-only model with a three-epoch decoder LoRA. Fourteen global tokens are better than eight at every k, so we use $g = 14$.

Computational cost. We distinguish three kinds of cost. *Decoder input*: ViBridge-VQA passes 50 visual tokens to the decoder, against up to 1,792 for the fine-tuned Vintern-1B. *Vision encoding*: InternViT-300M processes 577 tokens for one 336×336 view, about 191 GMACs, against 1,025 tokens and 362 GMACs for each 448×448 view (analytic counts, matching fvcore at 448×448); with six tiles and a thumbnail, the fine-tuned Vintern-1B thus needs up to 2,533 GMACs per image, 13 times more. *Evaluation runtime*: within our own budgets, keeping all 144 local tokens instead of 36 makes an evaluation session about 17% longer (Table 2).

4.5 Ablation Study

Where should the visual tokens come from? Table 3 compares three sources of visual tokens for the frozen decoder. The first group reads only the global `[CLS]` vector, through a *Residual* bridge that produces one token or the *Multi-Token* bridge of Eq. (1). The second group learns to compress the patch tokens,

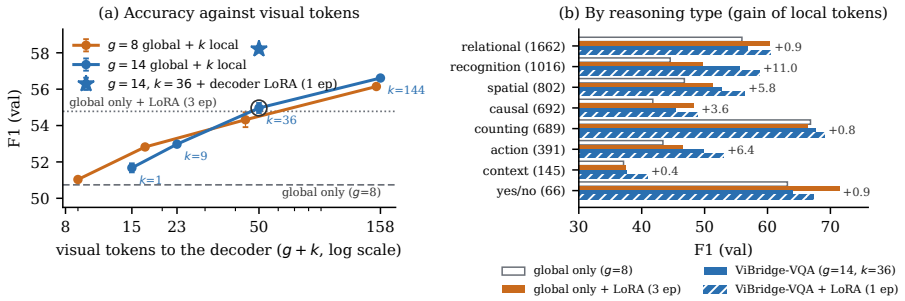


Fig. 2: (a) Validation F1 against the number of visual tokens (mean ± std over three seeds); the circled point is the knee, $g = 14$, $k = 36$. Horizontal lines: the global-only model (four seeds) and the same model with a three-epoch decoder LoRA (three seeds). (b) Validation F1 by reasoning type (number of questions in parentheses), each model averaged over all of its seeds (global only: four; others: three); the numbers give the gain of ViBridge-VQA ($g = 14$, $k = 36$, no LoRA) over the global-only model ($g = 8$).

either with a *Tile-Attention* bridge in which eight learned queries attend over the patches, or with a light or full Q-Former [11]. Without decoder adaptation, the Multi-Token bridge is the best of these five, and the bridges that compress the patches are 3.7–5.6 F1 lower despite having up to 9.4× more parameters. Learning a new alignment of the patches from 26k questions does not pay off. ViBridge-VQA, which takes the local tokens from the pre-trained projector instead, improves over the best global-only bridge by 4.2 F1 with $k = 36$ and by 5.9 F1 with $k = 144$.

Local tokens and decoder adaptation are complementary. A decoder LoRA improves every global-only bridge and brings them within 0.8 F1 of one another (52.69–53.47), which suggests that the decoder learns to compensate for poor visual input. Even after this adaptation, all of them remain below ViBridge-VQA with a frozen decoder. Decoder adaptation is therefore not needed to surpass them, but it is still useful: in the same one-epoch joint training, it adds 3.3 F1 to ViBridge-VQA, more than the 2.7 F1 it adds to the Multi-Token bridge. Conversely, distilling the projector into the global-only bridge, at the feature or output level, lowers F1 by 0.3–0.4 points: the projector's information is best passed as tokens.

4.6 Analysis by Reasoning Type and Error Analysis

Reasoning types. Fig. 2b breaks the results down by reasoning category. Local tokens help most where the answer depends on what is visible in the image: recognition (+11.0 F1 over the global-only model, Fig. 2b), action (+6.4) and spatial (+5.8) questions. On these categories, ViBridge-VQA with a frozen decoder is also better than the global-only model with a decoder LoRA. Conversely, decoder adaptation is more useful for causal and relational questions, whose

Table 3: Where the visual tokens come from, with and without decoder adaptation (validation, ×100, mean over three seeds with ± std for F1; Multi-Token: four seeds; *seed 42 only). Bridge only: two epochs; + LoRA: rank-16 decoder LoRA (2.16M) trained jointly for one epoch. Params: bridge parameters.

			Bridge only		+ decoder LoRA	
Visual tokens from	Tokens	Params	F1	CIDEr	F1	CIDEr
Global vector (`[CLS]`)						
Residual	1	4.86M	46.02 ±.36	86.48	52.69 ±.14	104.25
Multi-Token	8	7.35M	50.74 ±.17	99.07	53.47 ±.17	106.68
Learned compression of the patch tokens						
Tile-Attention	8	4.14M	45.14 ±.92	84.14	52.82*	104.72*
Light Q-Former	8	27.6M	47.06 ±.66	88.07	53.21 ±.14	106.26
Full Q-Former	16	69.4M	46.84 ±.47	87.74	53.29 ±.29	106.50
Global vector + pre-trained projector (ViBridge-VQA, $g = 14$)						
$k = 36$	50	12.86M	54.96 ±.28	113.29	**58.21** ±.16	**124.88**
$k = 144$	158	12.86M	**56.61** ±.04	**120.71**	–	–

reference answers are the longest (5.0 and 5.6 words on average), and for yes/no questions. Adding the LoRA to ViBridge-VQA combines both strengths: the resulting model is the best of the four models in Fig. 2b in seven of the eight categories, matching the global-only model with LoRA on causal (48.9 versus 48.3 F1) and relational (60.5 versus 60.4) questions while keeping the large gain of the local tokens on recognition (58.7 versus 49.6).

Error analysis. An analysis of our validation predictions (all seeds) helps explain the low exact-match accuracy of all systems. Our answers are as long as the references (4.3 words on average). Most predictions (77–81%) share some, but not all, words with the closest reference: a correct answer is often phrased differently from all five references. Local tokens turn some of these partial answers into exact ones and reduce complete failures: compared with the global-only model, ViBridge-VQA increases the share of predictions with F1 = 1 from 9.7% to 13.3% and reduces the share with no word in common with any reference from 9.6% to 8.0% (6.8% with LoRA). Complete failures concentrate on context, recognition, spatial and action questions (12–18%), i.e., on fine-grained perception, and are rare (< 5%) for counting, causal and relational questions.

4.7 Generalisation to Other Datasets

Table 4 evaluates the models without further training on four Vietnamese VQA datasets: two scene-text datasets, ViTextVQA [22] and OpenViVQA [20], and two general-domain ones, ViVQA-X [7] and ViVQA [27]. The strongest global-only model (with a three-epoch LoRA) serves as a reference. Local tokens change the picture most on scene text, where the global-only model collapses, presumably because the `[CLS]` vector cannot convey printed words. On ViTextVQA, F1 rises from 3.96 to 15.14 with 36 local tokens and to 33.26 with 144, slightly above the zero-shot Vintern-1B that reads six 448×448 tiles, although ViBridge-VQA reads a single 336×336 view. On OpenViVQA, whose references are long descriptive sentences, local tokens narrow the gap to the six-tile Vintern-1B.

Table 4: Zero-shot transfer to Vietnamese VQA datasets not seen during training (×100; about 1,000 sampled test questions per sampling seed, 505–519 for ViVQA; mean over three sampling seeds, ± std for F1). Vintern-1B reads six tiles and a thumbnail; the other models are trained on AutoViVQA only (seed 42). Best results per column are in bold.

	ViTextVQA		OpenViVQA		ViVQA-X		ViVQA	
Model	F1	CIDEr	F1	CIDEr	F1	CIDEr	F1	CIDEr
Vintern-1B, zero-shot	32.01 ±.31	**195.62**	**58.40** ±.80	**411.63**	14.52 ±.47	50.13	23.90 ±.24	78.63
Global only + LoRA, 3 ep.	3.96 ±.09	15.64	21.56 ±.48	83.21	15.89 ±.65	42.12	31.48 ±1.15	84.94
ViBridge-VQA, k=36	15.14 ±.47	66.39	28.24 ±.79	128.36	24.91 ±.14	79.51	42.68 ±1.52	171.21
ViBridge-VQA, k=144	**33.26** ±.67	190.30	35.89 ±.80	187.00	**27.89** ±.13	92.22	**44.95** ±.90	**191.28**
ViBridge-VQA, k=36 + LoRA	15.39 ±.39	70.07	28.99 ±.51	130.46	27.64 ±.11	**95.29**	44.76 ±1.21	183.60

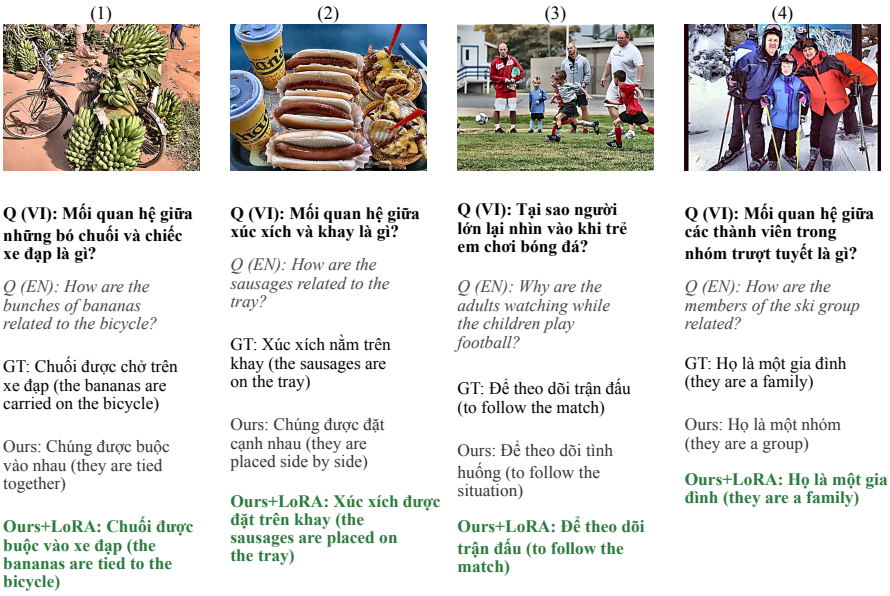


Fig. 3: Qualitative examples from the AutoViVQA validation set (seed 42). Q: question, GT: one of the five reference answers; Ours: ViBridge-VQA ($g = 14$, $k = 36$), Ours+LoRA: ViBridge-VQA with a one-epoch decoder LoRA.

On the general-domain datasets, ViBridge-VQA beats both references, by 10.4 (ViVQA-X) and 18.8 (ViVQA) F1 over Vintern-1B with $k = 36$. The zero-shot model again pairs high recall (61.2 on ViVQA-X) with very low precision (9.1). Adding the decoder LoRA to ViBridge-VQA helps on these general-domain datasets, bringing $k = 36$ close to $k = 144$, but barely on scene text. Reading text thus requires more local tokens, which decoder adaptation cannot replace.

4.8 Qualitative Analysis

Fig. 3 illustrates how the two components work together. In every example, ViBridge-VQA already identifies the relevant content of the image (the bananas and the bicycle, the sausages, the football match, the skiers), but its answer remains vague. Adding the decoder LoRA turns it into the precise relation or reason that the question asks for, such as the bananas being tied *to the bicycle* or the skiers being *a family*, in line with the gains of decoder adaptation on relational and causal questions (Section 4.6).

5 Conclusion

In this work, we introduced **ViBridge-VQA**, a parameter-efficient approach that improves the Vietnamese VLM Vintern-1B for generative VQA without updating any of its pre-trained weights, by giving its frozen decoder trainable global tokens from the `[CLS]` vector and local tokens from the pre-trained projector. With 14 global and 36 local tokens, the knee of the accuracy–cost Pareto front on validation, training only the bridge raises F1 on AutoViVQA from 17.6 to 54.6, above the published fine-tuned Vintern-1B on seven of eight metrics, and training a lightweight decoder LoRA with it raises F1 to 58.2. Local tokens matter far more than global ones and complement decoder adaptation.

The task-specific ViMoE-VQA still reaches a higher F1, and accuracy had not saturated at 144 local tokens, which suggests that higher-resolution inputs could narrow this gap and further improve text reading, for instance on OpenViVQA.

Overall, an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and how many, with an explicit accuracy–cost criterion. The code and the grouped split are available at `https://github.com/nguyennn263/modular-vlm-finetune`. We hope that this work will encourage efficient and carefully evaluated multimodal models for low-resource languages.

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References

1. Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: a visual language model for few-shot learning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 23716–23736 (2022)
2. Banerjee, S., Lavie, A.: METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72 (2005)
3. Cai, M., Yang, J., Gao, J., Lee, Y.J.: Matryoshka multimodal models (2024), arXiv preprint arXiv:2405.17430

4. Cha, J., Kang, W., Mun, J., Roh, B.: Honeybee: Locality-enhanced projector for multimodal LLM. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13817–13827 (2024)
5. Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., Li, B., Luo, P., Lu, T., Qiao, Y., Dai, J.: InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 24185–24198 (2024)
6. Doan, K.T., Huynh, B.G., Hoang, D.T., Pham, T.D., Pham, N.H., Nguyen, Q.T.M., Vo, B.Q., Hoang, S.N.: Vintern-1B: An efficient multimodal large language model for Vietnamese (2024), arXiv preprint arXiv:2408.12480
7. Duong, T.B., Tran, H.M., Le-Nguyen, B.N., Duong, D.T.: An automated pipeline for constructing a Vietnamese VQA-NLE dataset. In: Proceedings of the International Conference on Intelligent Systems and Networks (ICISN). pp. 164–173. Lecture Notes in Networks and Systems, Springer (2026)
8. Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 2790–2799 (2019)
9. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2022)
10. Hwang, C.L., Yoon, K.: Multiple Attribute Decision Making: Methods and Applications. Springer (1981)
11. Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 19730–19742 (2023)
12. Li, K.Y., Goyal, S., Semedo, J.D., Kolter, J.Z.: Inference optimal VLMs need only one visual token but larger models (2024), arXiv preprint arXiv:2411.03312
13. Li, X.L., Liang, P.: Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 4582–4597 (2021)
14. Lin, C.Y.: ROUGE: A package for automatic evaluation of summaries. In: Proceedings of the ACL Workshop on Text Summarization Branches Out. pp. 74–81 (2004)
15. Lin, T.Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Zitnick, C.L., Dollár, P.: Microsoft COCO: Common objects in context. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 740–755 (2014)
16. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 26296–26306 (2024)
17. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 34892–34916 (2023)
18. Nguyen, A.T., Nguyen, T.Q., Dang, D.L., Tran, D.T., Phan, B.D., Nguyen, Q.T., Le, T.: ViMoE-VQA: Reasoning-aware sparse experts for generative Vietnamese visual question answering. Procedia Computer Science (2026), to appear
19. Nguyen, A.T., Phan, B.D., Nguyen, T.Q., Tran, D.T., Dang, D.L., Nguyen, Q.T., Le, T.: AutoViVQA: A large-scale, automatically constructed dataset for Vietnamese

visual question answering. In: Proceedings of the Asian Conference on Intelligent Information and Database Systems (ACIIDS). pp. 492–506. Communications in Computer and Information Science, Springer (2026)
20. Nguyen, N.H., Vo, D.T.D., Nguyen, K.V., Nguyen, N.L.T.: OpenViVQA: Task, dataset, and multimodal fusion models for visual question answering in Vietnamese. Information Fusion **100**, 101868 (2023)
21. Nguyen, N.S., Nguyen, V.S., Le, T.: Advancing Vietnamese visual question answering with transformer and convolutional integration. Computers and Electrical Engineering **119**, 109474 (2024)
22. Nguyen, Q.V., Tran, D.Q., Pham, H.Q., Nguyen, T.K.B., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViTextVQA: A large-scale visual question answering dataset for evaluating Vietnamese text comprehension in images (2025), arXiv preprint arXiv:2404.10652
23. Papineni, K., Roukos, S., Ward, T., Zhu, W.J.: BLEU: a method for automatic evaluation of machine translation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 311–318 (2002)
24. Pham, H.Q., Nguyen, T.K.B., Nguyen, Q.V., Tran, D.Q., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViOCRVQA: Novel benchmark dataset and vision reader for visual question answering by understanding Vietnamese text in images (2024), arXiv preprint arXiv:2404.18397
25. Qwen Team: Qwen2.5 technical report (2024), arXiv preprint arXiv:2412.15115
26. Satopää, V., Albrecht, J., Irwin, D., Raghavan, B.: Finding a "kneedle" in a haystack: Detecting knee points in system behavior. In: Proceedings of the International Conference on Distributed Computing Systems Workshops (ICDCSW). pp. 166–171 (2011)
27. Tran, K.Q., Nguyen, A.T., Le, A.T.H., Nguyen, K.V.: ViVQA: Vietnamese visual question answering. In: Proceedings of the Pacific Asia Conference on Language, Information and Computation (PACLIC). pp. 683–691 (2021)
28. Tran, K.V., Nguyen, K.V., Nguyen, N.L.T.: BARTPhoBEiT: Pre-trained sequence-to-sequence and image transformers models for Vietnamese visual question answering. In: Proceedings of the International Conference on Multimedia Analysis and Pattern Recognition (MAPR). pp. 1–6 (2023)
29. Tran, K.V., Phan, H.P., Nguyen, K.V., Nguyen, N.L.T.: ViCLEVR: A visual reasoning dataset and hybrid multimodal fusion model for visual question answering in Vietnamese (2023), arXiv preprint arXiv:2310.18046
30. Tsimpoukelli, M., Menick, J., Cabi, S., Eslami, S.M.A., Vinyals, O., Hill, F.: Multimodal few-shot learning with frozen language models. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 200–212 (2021)
31. Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4566–4575 (2015)